

Timothy Do's Full Resume

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EDUCATION

University of California Irvine, Irvine, CA — *B.S. Electrical Engineering*

September 2019 - June 2023

- 2nd Year Undergraduate, Junior Standing
- Overall GPA: 3.97, Major GPA: 4.0
- **Specialization:** Digital Signal Processing
- **Research Interests:** Digital Signal Processing with Artificial Intelligence, Image Enhancements, Motion Interpolation, and Pixel Manipulation to make media more enjoyable.
- **Relevant Coursework:**
 - EECS 10: An Introduction to C Programming, learned about compilers, data structures, and linux commands (Fall 2019)
 - EECS 31: An Introduction to Digital Logic, learned about logic gates, boolean algebra, and system design. (Winter 2020)
 - EECS 31L: Digital Logic Lab, experience with VHDL, applying knowledge from EECS 31 into applications. (Spring 2020)
 - Math 2D: Multivariable Calculus, learned about partial differentiation, vectors, and double integrals (Fall 2020)
 - Math 3A: Linear Algebra, learned about column spaces, matrix operations, and eigenvalues/eigenvectors (Winter 2020)
 - Math 3D: Differential Equation, learned about solving ODEs using integrating factor, eigenvector method, Power Series representation, and Laplace Transforms (Spring 2020)
- **Major Coursework** for Fall 2020: Math 2E (Multivariable Calculus), Physics 7E (Fluids and Waves), and EECS 20 (Assembly Language and C)
- **Awards:** Dean's Honor List (GPA 3.5+) for 3 Quarters (Fall 2019, Winter 2020, and Spring 2020)

Evergreen Valley High School, San Jose, CA — *Diploma*

August 2015 - May 2019

- GPA: Unweighted: 3.8, Weighted: 4.0
- **Relevant Coursework:** AP Physics C: Mechanics (4), AP Computer Science A (4), AP Statistics (4), AP Calculus AB (5), AP US History (4), Math 072 (Calc II) @ Evergreen Valley College
- **Awards:** EVHS Band Scholar, Magna Cum Laude, Golden State Diploma, Bilingual Certification in Vietnamese, AP Scholar with Distinction

SKILLS

Web Programming - Experience with HTML, CSS, and Javascript and managing websites.

Compiler / Object-Oriented Programming - Experience with C, Java, and Python libraries

GPIO Circuitry/Programming: Programming GPIO pins on the Raspberry PI using Python and the GPIO Library. Also programmed Arduino circuits in C.

Photo/Video Editing - Experience with editing programs such as Final Cut Pro, Photoshop, & Premiere Pro.

Computers - Experience with circuit board design and assembly for Desktop machines.

Music - Experience with Musical Composition, played 4 instruments including Euphonium & Trombone

ACTIVITIES

Github Arctic Vault Contributor

Principal Euphonium for the 2019 Evergreen Valley Wind Ensemble

3rd Chair Euphonium Player for the 2019 Santa Clara County Honor Band

Former Euphonist for EVYO (Evergreen Valley Youth Orchestra)

Teacher's Assistant for Most Holy Trinity Parish

LANGUAGES

English (Fluent), **Vietnamese** (Fluent), & **Spanish** (Proficient)

EXPERIENCE

Boy Scouts of America Troop 610, San Jose, CA — *Eagle Scout*

September 2013 - September 2019

- Mentored youth between 12-17 on the principles of scouting
- Awarded the Bronze Palm: Earning 5+ additional Merit Badges over the 21 required for Eagle Scout.
- **Eagle Scout Project:**
 - Constructing 2 Food Cabinets for the Milpitas Food Bank
 - Required detailed planning, sketches, and team management
- **Troop Roles:**
 - Bugler: Played the trumpet/Euphonium during troop outings
 - ASPL: Managed troop as a whole with their daily activities

Evergreen Music Mentors, San Jose, CA — *Founding President*

September 2018 - Present

- Lead music tutoring sessions for younger middle school students
- Managed private tutoring sessions with employees and clients
- Managed financial transactions for the company
- Integrated software (Cybernotes) as a video calling platform

Environmental Injustice , Irvine, CA — *Collaborator*

June 2020 - Present

- Admin for the discord server, managing working groups
- Managed Instagram account, creating captions/visuals using Canva.
- Collaborated with 11 International Team Members remotely to research on chemical disasters, pollution, and climate change
- Developed techniques such as stakeholder analysis, using Civic Data Tools such as CalEnviroScreen , and citing in Chicago format.

PUBLICATIONS

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Fast Disaster Case Study: Santa Fe Springs](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Slow Disaster Case Study: Wilmington](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine.

Tahis Alcantar; **Timothy Do**; April Godinez; Daniel Jilani; Vicent Marin; Wonhee Lee; Haoyang Liu; Huiqi Mai; Andrew Ramirez; Jiaqi Wu; Depei Xu. 2020. [Combo Disaster Case Study: San Bernardino County](#). UCI Anthropology 25A, Environmental Injustice. Disaster STS Network. University of California Irvine

PROJECTS

[RefreshVideos](#) — *A High Refresh Rate Video Player*

April 2020 - Present

- **Languages Used:** HTML / CSS / Javascript
- Used Canvas to play videos at high frame rates and apply effects
- Featuring filters like Edge Detection, Aging, etc.

[PiFmMorse](#) — *A Morse Code FM Transmitter for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to transmit Morse Code over FM radio via GPIO 4
- Learned to concatenate .wav files via wave library
- Learned to call linux commands using os library

[RPiMorse](#) — *A Morse Code LED Circuit for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to flash an LED in Morse Code given a circuit
- Learned GPIO Pins Wiring and its respective python library

[Timduino Tutorials](#) — *Experiments with the Arduino Uno*

September 2020

- **Languages Used:** C
- Learned functions to set voltages (pinMode(), digitalWrite())
- Wired Circuits and pins to apply Arduino Programs

[SCU Complex Fires](#) — *A Website Providing Wildfire Resources*

August 2020 - September 2020

- **Languages Used:** HTML / CSS/ Javascript
- Provided Wildfire resources for research, shelter, and news
- Worked with a team of 4 and remotely collaborated through Github
- Features an Energy-Efficient Dark Mode.

[Cybernotes](#) — *An Online Web App for Musicians*

July 2019

- **Languages Used:** HTML / CSS / Javascript / node.js
- Features video calling, live sheet music viewing via Google Drive, & a tuner/metronome module

[EVHub](#) — *An App for Evergreen Valley High School*

May 2019 - May 2019

- **Languages Used:** Java
- App meant to help high school students manage their coursework, with features such as social media responses, school info, etc.
- Work as a group of 4 and collaborated in Github
- First repository, learned how to use git